Adjacency List Representation Displayed on Output

```
> java GraphLists
Parts[] = 13 22
Reading edges from text file
Edge A--(2)--F
Edge A--(1)--B
Edge A--(6)--G
Edge B--(2)--D
Edge B--(4)--E
Edge B--(1)--C
Edge C--(4)--E
Edge D--(2)--E
Edge E--(2)--F
Edge E--(4)--L
Edge E--(1)--G
Edge F--(1)--D
Edge F--(2)--L
Edge G--(5)--L
Edge G--(1)--J
Edge G--(3)--H
Edge H--(2)--I
Edge I--(1)--K
Edge J--(1)--K
Edge J--(3)--L
Edge J--(2)--M
Edge L--(1)--M

adj[A] -> |G | 6| -> |B | 1| -> |F | 2| ->
adj[B] -> |C | 1| -> |E | 4| -> |D | 2| -> |A | 1| ->
adj[C] -> |E | 4| -> |B | 1| ->
adj[D] -> |F | 1| -> |E | 2| -> |B | 2| ->
adj[E] -> |G | 1| -> |L | 4| -> |F | 2| -> |D | 2| -> |C | 4| -> |B | 4| ->
adj[F] -> |L | 2| -> |D | 1| -> |E | 2| -> |A | 2| ->
adj[G] -> |H | 3| -> |J | 1| -> |L | 5| -> |E | 1| -> |A | 6| ->
adj[H] -> |I | 2| -> |G | 3| ->
adj[I] -> |K | 1| -> |H | 2| ->
adj[J] -> |M | 2| -> |L | 3| -> |K | 1| -> |G | 1| ->
adj[K] -> |J | 1| -> |I | 1| ->
adj[L] -> |M | 1| -> |J | 3| -> |G | 5| -> |F | 2| -> |E | 4| ->
adj[M] -> |L | 1| -> |J | 2| ->
```

Breadth First Search Output

```
Visited Initial Vertex A
Inserting initial vertex into queue

Exploring: A

Found A->G
Inserting G into queue
Current State of Queue:
G->
Found A->B
Inserting B into queue
Current State of Queue:
G->B->
Found A->F
Inserting F into queue
Current State of Queue:
G->B->F->

Exploring: G

Found G->H
Inserting H into queue
Current State of Queue:
B->F->H->
Found G->J
Inserting J into queue
Current State of Queue:
B->F->H->J->
Found G->L
Inserting L into queue
Current State of Queue:
B->F->H->J->L->
Found G->E
Inserting E into queue
Current State of Queue:
B->F->H->J->L->E->
Found G->A
Current State of Queue:
B->F->H->J->L->E->
```

```
Exploring: B

Found B->C
Inserting C into queue
Current State of Queue:
F->H->J->L->E->C->
Found B->E
Current State of Queue:
F->H->J->L->E->C->
Found B->D
Inserting D into queue
Current State of Queue:
F->H->J->L->E->C->D->
Found B->A
Current State of Queue:
F->H->J->L->E->C->D->

Exploring: F

Found F->L
Current State of Queue:
H->J->L->E->C->D->
Found F->D
Current State of Queue:
H->J->L->E->C->D->
Found F->E
Current State of Queue:
H->J->L->E->C->D->
Found F->A
Current State of Queue:
H->J->L->E->C->D->

Exploring: H

Found H->I
Inserting I into queue
Current State of Queue:
J->L->E->C->D->I->
Found H->G
Current State of Queue:
J->L->E->C->D->I->
```

```
Exploring: J

Found J->M
Inserting M into queue
Current State of Queue:
L->E->C->D->I->M->
Found J->L
Current State of Queue:
L->E->C->D->I->M->
Found J->K
Inserting K into queue
Current State of Queue:
L->E->C->D->I->M->K->
Found J->G
Current State of Queue:
L->E->C->D->I->M->K->

Exploring: L

Found L->M
Current State of Queue:
E->C->D->I->M->K->
Found L->J
Current State of Queue:
E->C->D->I->M->K->
Found L->G
Current State of Queue:
E->C->D->I->M->K->
Found L->F
Current State of Queue:
E->C->D->I->M->K->
Found L->E
Current State of Queue:
E->C->D->I->M->K->
```

```
Exploring: E

Found E->G
Current State of Queue:
C->D->I->M->K->
Found E->L
Current State of Queue:
C->D->I->M->K->
Found E->F
Current State of Queue:
C->D->I->M->K->
Found E->D
Current State of Queue:
C->D->I->M->K->
Found E->C
Current State of Queue:
C->D->I->M->K->
Found E->B
Current State of Queue:
C->D->I->M->K->

Exploring: C

Found C->E
Current State of Queue:
D->I->M->K->
Found C->B
Current State of Queue:
D->I->M->K->

Exploring: D

Found D->F
Current State of Queue:
I->M->K->
Found D->E
Current State of Queue:
I->M->K->
Found D->B
Current State of Queue:
I->M->K->
```

```
Exploring: I

Found I->K
Current State of Queue:
M->K->
Found I->H
Current State of Queue:
M->K->

Exploring: M

Found M->L
Current State of Queue:
K->
Found M->J
Current State of Queue:
K->

Exploring: K

Found K->J
Current State of Queue:

Found K->I
Current State of Queue:
```

Depth First Search Output

```
Visited: A

Exploring A

Found A->G
Going from A to G
Visited: G

Exploring G

Found G->H
Going from G to H
Visited: H

Exploring H

Found H->I
Going from H to I
Visited: I

Exploring I

Found I->K
Going from I to K
Visited: K

Exploring K

Found K->J
Going from K to J
Visited: J

Exploring J

Found J->M
Going from J to M
Visited: M

Exploring M

Found M->L
Going from M to L
Visited: L

Exploring L

Found L->M
M Already Visited
```

```
Exploring L

Found L->J
J Already Visited

Exploring L

Found L->G
G Already Visited

Exploring L

Found L->F
Going from L to F
Visited: F

Exploring F

Found F->L
L Already Visited

Exploring F

Found F->D
Going from F to D
Visited: D

Exploring D

Found D->F
F Already Visited

Exploring D

Found D->E
Going from D to E
Visited: E

Exploring E

Found E->G
G Already Visited

Exploring E

Found E->L
L Already Visited
```

```
Exploring E

Found E->F
F Already Visited

Exploring E

Found E->D
D Already Visited

Exploring E

Found E->C
Going from E to C
Visited: C

Exploring C

Found C->E
E Already Visited

Exploring C

Found C->B
Going from C to B
Visited: B

Exploring B

Found B->C
C Already Visited

Exploring B

Found B->E
E Already Visited

Exploring B

Found B->D
D Already Visited

Exploring B

Found B->A
A Already Visited
Going back to C from B
Going back to E from C
```

```
Exploring E

Found E->B
B Already Visited
Going back to D from E

Exploring D

Found D->B
B Already Visited
Going back to F from D

Exploring F

Found F->E
E Already Visited

Exploring F

Found F->A
A Already Visited
Going back to L from F

Exploring L

Found L->E
E Already Visited
Going back to M from L

Exploring M

Found M->J
J Already Visited
Going back to J from M

Exploring J

Found J->L
L Already Visited

Exploring J

Found J->K
K Already Visited

Exploring J

Found J->G
G Already Visited
Going back to K from J
```

```
Exploring K

Found K->I
I Already Visited
Going back to I from K

Exploring I

Found I->H
H Already Visited
Going back to H from I

Exploring H

Found H->G
G Already Visited
Going back to G from H

Exploring G

Found G->J
J Already Visited

Exploring G

Found G->L
L Already Visited

Exploring G

Found G->E
E Already Visited

Exploring G

Found G->A
A Already Visited
Going back to A from G

Exploring A

Found A->B
B Already Visited

Exploring A

Found A->F
F Already Visited
```